Government of **Western Australia**
School Curriculum and Standards Authority

Western Australian Certificate of Education Examination, 2012

Question/Answer Booklet

EARTH AND ENVIRONMENTAL SCIENCE
Stage 3

Please place your student identification label in this box

Student Number: In figures

In words

Time allowed for this paper
Reading time before commencing work: ten minutes
Working time for paper: three hours

Materials required/recommended for this paper
To be provided by the supervisor
This Question/Answer Booklet
Multiple-choice Answer Sheet

Number of additional answer booklets used (if applicable):

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction tape/fluid, eraser, ruler, highlighters

Special items: protractor, drawing compass, mathomat, non-programmable calculators approved for use in the WACE examinations

Important note to candidates
No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Ref: 12-040

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of total exam
Section One: Multiple-choice	15	15	15	15	10
Section Two: Short answer	9	9	105	105	60
Section Three: Extended response	3	2	60	30	30
				Total	100

Instructions to candidates

1. The rules for the conduct of Western Australian external examinations are detailed in the *Year 12 Information Handbook 2012*. Sitting this examination implies that you agree to abide by these rules.

2. Answer the questions according to the following instructions.

   Section One: Answer all questions on the separate Multiple-choice Answer Sheet provided. For each question shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. If you make a mistake, place a cross through that square then shade your new answer. Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

   Sections Two and Three: Write your answers in this Question/Answer Booklet.

3. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.

4. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
   - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
   - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question that you are continuing to answer at the top of the page.

Section One: Multiple-choice **10% (15 Marks)**

This section has **15** questions. Answer **all** questions on the separate multiple-choice Answer Sheet provided.

For each question shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. If you make a mistake, place a cross through that square, then shade your new answer. Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Suggested working time: 15 minutes.

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1. A scientist studying a series of satellite photographs of a large Indonesian island taken between 1990 and 2010 noted a very rapid reduction in the amount of rain forest cover. He estimated that, in some parts, up to 60% of the original 1990 cover had been destroyed by 2010.

   Which one of the following statements **best** explains the destruction of the rain forest?

   (a) The loss of trees is related to a rapidly drying climate because of La Niña.
   (b) The loss of trees is related to rising levels of salt in the soil.
   (c) The loss of trees is related to human economic activity.
   (d) The loss of trees is related to holes developing in the ozone layer.

2. Long-term heating of the Earth's atmosphere over the past 200 years is probably caused by

   (a) a steady increase in the volume of dust produced by increased volcanic activity.
   (b) the rapid melting of polar ice caps and glaciers.
   (c) the heat released by the burning of forests by humans.
   (d) an increase in carbon dioxide (CO_2) related to the combustion of fossil fuels.

3. Which is the most likely effect of a **decrease** in the concentration of ozone due to the release of CFCs into the stratosphere?

   (a) increased amounts of ultra violet (UV) radiation reaching the Earth's surface
   (b) increased amounts of carbon dioxide (CO_2) being absorbed by the oceans
   (c) decreased amounts of oxygen in the troposphere
   (d) decreased rainfall in tropical regions

4. The **best** site for a new major waste disposal centre at which rubbish is to be buried would be a location

   (a) near major transport links where the natural vegetation has already been cleared.
   (b) where soft underlying rocks reduce the costs of excavation.
   (c) where the water table is close to the surface and it is unsuitable for housing.
   (d) where water cannot penetrate the underlying rocks.

5. Which one of the following statements **best** describes the conditions in the atmosphere that produce a localised sea breeze in places close to the coast?

(a) In summer, the pressure over the ocean is lower than the pressure over the land, so the air flows from the sea to the land.
(b) In summer, the pressure over the ocean is higher than pressure over the land, so the air flows from the sea to the land.
(c) In winter, the pressure over the ocean is lower than pressure over the land, so the air flows from the sea to the land.
(d) In winter, the pressure over the ocean is higher than pressure over the land, so the air flows from the sea to the land.

6. The main process that heats the Earth's atmosphere is which one of the following?

(a) Incoming solar energy is trapped in the ozone layer.
(b) Incoming solar energy is absorbed by the atmosphere and clouds before reaching the surface.
(c) Energy is transferred from the Earth's surface to the atmosphere.
(d) Energy is radiated to space from the clouds and atmosphere.

7. Nutrients such as nitrogen-based fertilisers dissolve in water can build up in lakes, swamps and streams leading to a rapid growth of aquatic plants. This build-up of nutrients is called

(a) bioaccumulation.
(b) nitrification.
(c) acidification.
(d) eutrophication.

8. Which one of the following factors is the dominant cause of metamorphic foliation?

(a) chemical gradients
(b) temperature change
(c) high fluid content
(d) differential stress

9. What property of rocks is measured by geophysical gravity surveys?

(a) age
(b) magnetic susceptibility
(c) density
(d) electrical conductivity

10. Imagine that a geological map predicts the presence of limestone at Point 'A'. On travelling to that location, however, you find an exposure of granite instead. Which one of the following statements **best** describes the scientific implications of this discovery?

(a) Granite is an intrusive igneous rock with a high silica content.
(b) The predicted distribution of rocks is incorrect.
(c) Limestone can be metamorphosed into granite over time.
(d) Navigation in wilderness areas is often challenging.

See next page

11. Which one of the following processes is the dominant source of ozone in the stratosphere?

(a) the reaction between carbon dioxide and water during photosynthesis
(b) the destruction of atmospheric oxygen due to high levels of chlorofluorocarbons (CFCs)
(c) the reaction between oxygen molecules and oxygen atoms in the atmosphere
(d) the slow build-up of ozone from gases released during volcanic eruptions

12. Earth's global energy budget consists of the sum total gains of incoming energy and the total losses of outgoing energy. The sum of the gains is approximately equal to the sum of the losses, resulting in overall energy equilibrium. Which one of the following is a source of outgoing energy?

(a) solar radiation that is not reflected by clouds and the atmosphere
(b) geothermal energy produced by radioactive decay within the Earth's interior
(c) tidal energy produced by the gravitational interaction of the Earth, Moon and Sun
(d) long-wave radiation emitted to space by the atmosphere

13. A student collected and examined four metamorphic rock samples. Descriptions of each rock are shown below. Which one of the rocks is **most** likely gneiss?

(a) a white rock showing no foliation that fizzes when acid is added
(b) a rock showing alternating bands of light and dark minerals
(c) a dark shiny rock showing strongly foliated biotite and occasional large garnet crystals
(d) a dark fine grained rock that splits easily in large sheets

14. Which one of the following is **not** usually considered a source of global biomass?

(a) a coal deposit used to provide fuel for a power station
(b) solid wastes from a metropolitan sewerage system
(c) a pine plantation grown to produce furniture
(d) grain grown to make alcohol for use as a fuel

15. Which one of the following is the result of thermal convection within the oceans?

(a) stratification of the Swan River resulting from salinity differences
(b) the extreme tidal ranges found in northern Western Australia
(c) ocean currents such as the Leeuwin current off Perth
(d) the large ocean swells that can occur along the coast near Margaret River

End of Section One

See next page

Section Two: Short answer **60% (105 Marks)**

This section has **nine (9)** questions. Answer **all** questions. Write your answers in the spaces provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Suggested working time: 105 minutes.

Question 16 **(9 marks)**

The carbon cycle is the biogeochemical cycle by which carbon, stored in the biosphere, geosphere, hydrosphere and atmosphere of the Earth, is transferred between these reservoirs.

(a) Describe **one (1)** natural process that transfers carbon from the biosphere to the atmosphere. (3 marks)

(b) List **three (3)** ways in which human activity has altered the distribution of carbon in Earth's carbon cycle. (3 marks)

(c) Select **one (1)** of the three ways you listed in Part (b) and describe in detail how this human activity has altered the distribution of carbon. (3 marks)

Question 17 **(10 marks)**

'Acid rain' is the term used to describe precipitation (rain, snow, fog, dew) that is slightly acidic. Acid rain is formed when certain gases that are emitted into the atmosphere react with water molecules to form acids.

These gases are the by-products of human activity in regions where major industrial activity occurs and there are large urban populations.

(a) Name the **two (2)** gases most responsible for the production of acid rain. (2 marks)

(b) Describe how the release of each of the gases listed in Part (a) is linked to human activity. (4 marks)

(c) For each of the gases in Part (a) describe briefly **two (2)** measures that have been taken to reduce the emissions of the gases responsible for acid rain. (4 marks)

Question 18 **(11 marks)**

Intense volcanic activity on the island of Hawaii is seen as a consequence of the presence of an unusually hot mantle at the base of the lithosphere beneath the island – a phenomenon referred to as a 'mantle hot spot'.

(a) Describe, using an appropriately labelled diagram, how processes deep within the Earth generate and maintain such hot spot-related volcanism over geological time. (5 marks)

A series of extinct volcanoes extends to the north-west of this volcanically active region, forming a number of islands that decrease in size away from Hawaii.

(b) The age of the extinct volcanoes increases progressively with distance from Hawaii. What does this indicate about the relationship between mantle hot spots and lithospheric tectonic plates? (3 marks)

(c) The volcanic islands to the north-west of Hawaii decrease in size and eventually submerge with increasing distance from the volcanically active region. Explain this trend. (3 marks)

Question 19 **(11 marks)**

An ore is a type of rock that contains minerals with concentrations of valuable elements. An ore deposit is an economically significant accumulation of ore. Ores can be extracted through mining and then refined to extract the valuable elements. Western Australia has many world class ore deposits, particularly of metals.

Some ore deposits are formed by the redistribution of metals associated with hydrothermal activity.

(a) Using a labelled diagram, explain how hydrothermal mineralisation occurs. Indicate clearly the source of metals and fluids in the system you illustrate, and the physical conditions that are required for an ore deposit to form. (5 marks)

(b) Give an example of a Western Australian metallic ore deposit. Provide its approximate location and name the primary resource contained in this deposit. (3 marks)

(c) Name and describe an exploration technique that could be used to search for the mineral resource cited in Part (b) above. (3 marks)

Question 20 **(9 marks)**

The composition of the atmosphere has changed gradually during Earth's history. The earliest atmosphere, about 4.5 billion years ago, probably contained only hydrogen and helium. These light gases were quickly lost into space because Earth's gravity was too weak to retain them.

Volcanic eruptions release gases such as water vapour, carbon dioxide and sulfur dioxide. The atmosphere produced by early volcanic outgassing was thus very different to our current atmosphere. The composition of Earth's present atmosphere and of the likely earlier atmosphere is shown in the table below. Data for the earlier atmosphere are estimates based on the average percentages of gases currently released by volcanoes in a number of different tectonic environments.

Early volcanic outgassing atmosphere		**Current atmosphere**	
water vapour (H_2O)	66%	nitrogen (N_2)	78%
carbon dioxide (CO_2)	18%	oxygen (O_2)	21%
sulfur dioxide (SO_2)	12%	water vapour (H_2O)	0 – 4%
nitrogen (N_2)	trace	carbon dioxide (CO_2)	0.04%
oxygen (O_2)	nil	sulfur dioxide (SO_2)	trace
other	< 4%	other	<1%

(a) Our current atmosphere shows a significant drop in water vapour from that present during early volcanic outgassing. Explain why. (3 marks)

(b) The carbon dioxide content of the atmosphere decreased at the same time as the fall in the water vapour content described on page 14. Explain why. (3 marks)

(c) Oxygen gas is **not** released from volcanoes. Explain why there is now a large proportion of oxygen in Earth's atmosphere. (3 marks)

Question 21 **(14 marks)**

In the late 1970s, scientists observed a steady decline in the total volume of ozone in Earth's stratosphere (the ozone layer). This decline was particularly evident during spring over the Antarctic, where stratospheric ozone decline was severe enough to be referred to as an 'ozone hole'. The data below show the maximum extent of the stratospheric ozone hole over the Antarctic between 1979 and 2011.

Year*	Ozone hole area (million km^2)
1979	0.1
1985	14.2
1989	18.7
1993	24.2
1999	23.2
2003	25.8
2006	26.6
2011	24.7

* Note: Data are not given for all years.

(a) Draw a line graph to show these data on the grid provided. (5 marks)

(b) Suggest how human activity could have produced the ozone depletion displayed in the data. (3 marks)

A spare grid is provided on page 39. If you need to use it, cross out this attempt.

Question 21 (continued)

(c) Explain why the stratospheric ozone layer is important to life on this planet. (3 marks)

(d) High levels of tropospheric ozone have been implicated in a number of health problems. Describe **two (2)** human activities that can lead to abnormally high levels of ozone in the troposphere and name one specific chemical involved. (3 marks)

Question 22 **(9 marks)**

Australia is a large continent and shows great climatic variation, from the cool wet climates in Tasmania to the hot dry climates of the central regions and the tropical climates of the far north. Describe how each of the following can affect the climate of a region.

(a) Latitude (3 marks)

(b) Elevation (3 marks)

(c) Topography (3 marks)

See next page

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Question 23 **(16 marks)**

Igneous rocks can be named and classified by observing their texture and mineralogy. Common igneous rocks include basalt, dolerite, gabbro, andesite, diorite, rhyolite, pegmatite, granite, pumice, tuff and obsidian.

(a) During a field trip, a student recorded the following igneous rock descriptions. From the list of rocks above, suggest the most likely name of each rock. Classify each rock as mafic, felsic or intermediate. (6 marks)

a fine-grained igneous rock dominated by dark minerals

name ______________________ classification ______________________

a coarse-grained igneous rock containing quartz and feldspar with about 15% dark minerals

name ______________________ classification ______________________

a coarse-grained rock with approximately equal proportions of light and dark minerals

name ______________________ classification ______________________

(b) Obsidian and pumice are two igneous rocks produced by volcanic eruptions. Name and describe the distinctive textures of both. (4 marks)

Obsidian

____

____

____

____

Pumice

____

____

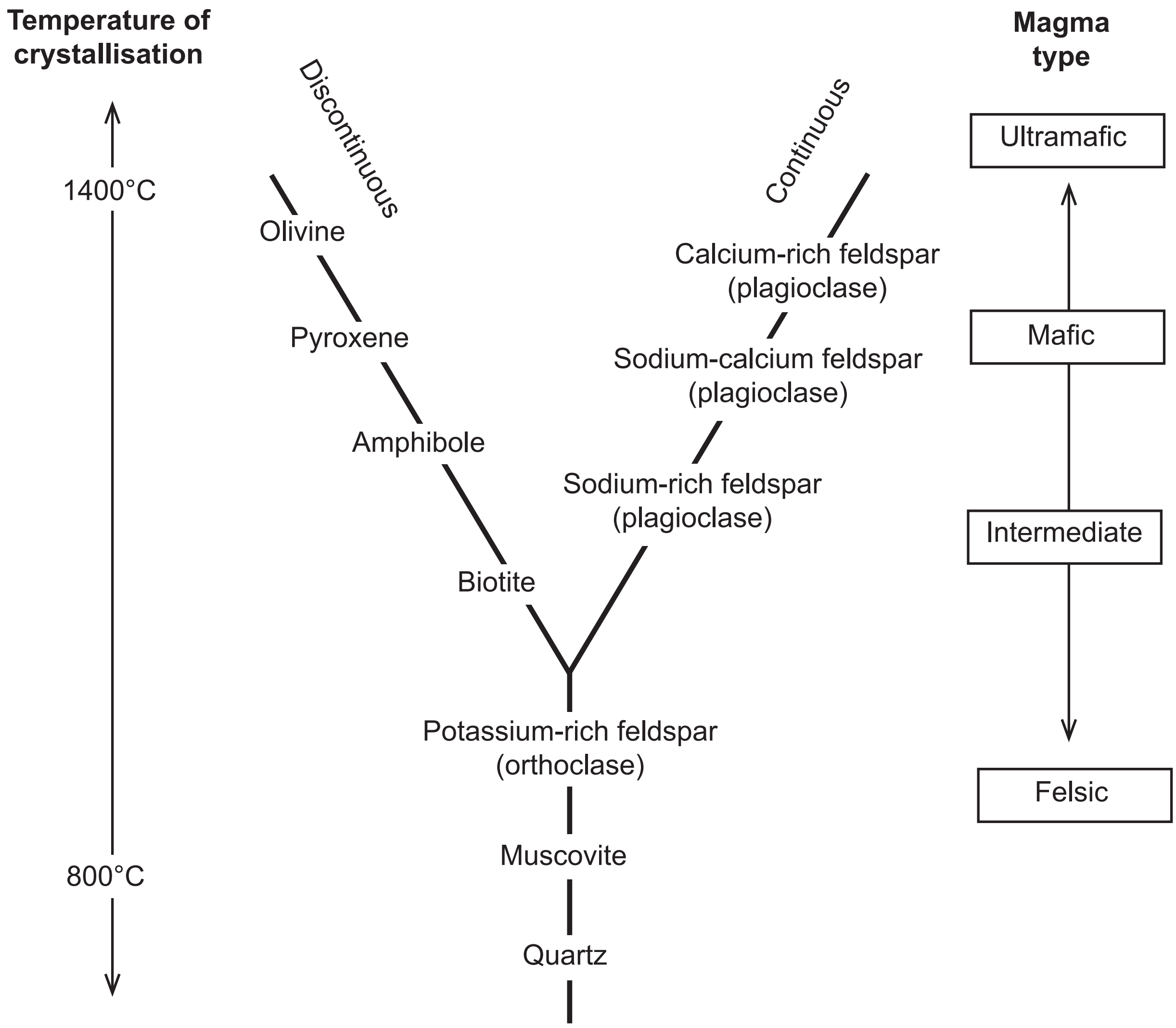
____

____

Question 23 (continued)

(c) In the early 1900s, the geologist Norman Bowen carried out a series of experiments to examine the process of crystallisation that occurs as igneous rocks form from magma. The illustration below is a summary of his findings, commonly referred to as Bowen's reaction series.

Bowen's reaction series



Use Bowen's reaction series and your knowledge of igneous processes to explain the following observations. (6 marks)

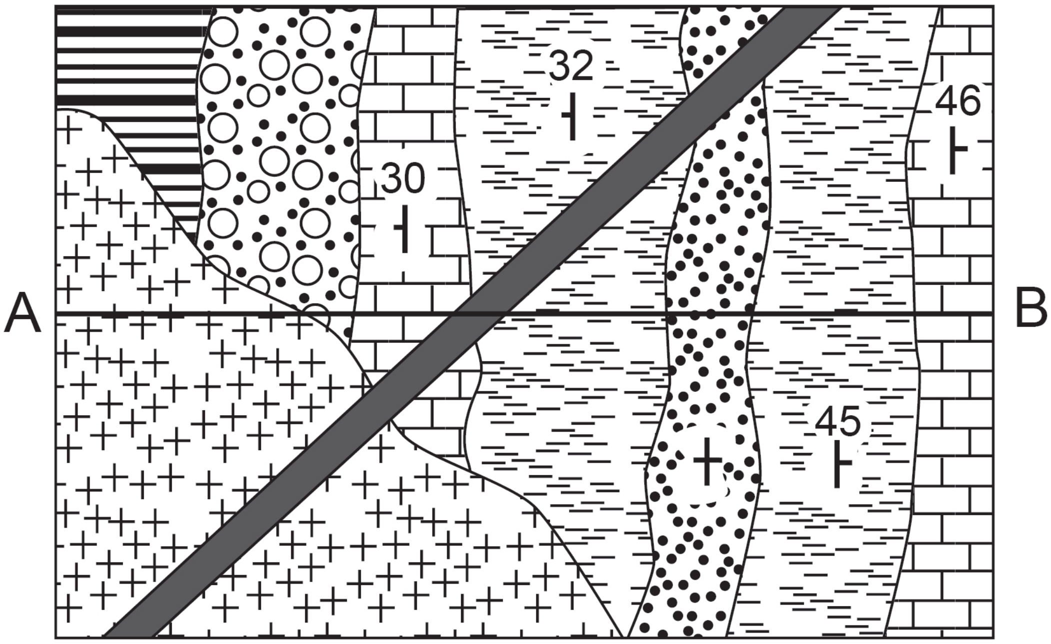
Amphibole in a granite often shows well-formed crystal faces, while quartz usually lacks crystal faces.

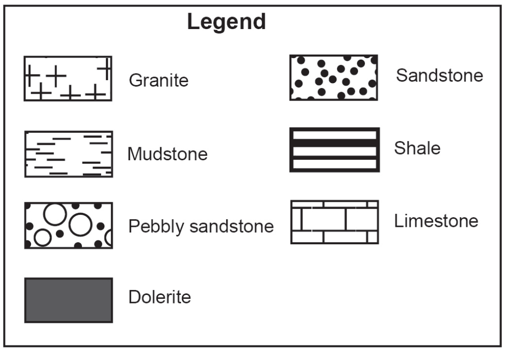
The plagioclase in basalts is usually calcium-rich, while the plagioclase in granites is usually sodium-rich.

Olivine is sometimes found in layers toward the base of a gabbro pluton.

Question 24 **(16 marks)**

The geological map below was produced by a student during a field trip.





(a) On the axes provided draw a geological cross-section from ‘A’ to ‘B’ to a depth of 400 m. Note: to assist you to transcribe strata locations, you may remove page 37 of the booklet by tearing along the perforations. The page **must** be reinserted into the booklet. (6 marks)

(b) Use the information shown and your cross-section drawn in Part (a) to complete the following: (4 marks)

Name the oldest rock unit shown on the cross-section.

Name the youngest rock unit shown.

Name the type of fold formed by the sedimentary layers.

Name the intrusive feature formed by the dolerite.

(c) Use the information shown on the map, your cross-section and your knowledge of natural processes to answer the following: (6 marks)

The student noticed that the granite was coarser grained toward the centre and became finer grained as he walked towards the edge. Suggest a reason for this observation.

Name the type of forces that produced the fold shown in the cross-section and suggest a tectonic environment in which the folding may have occurred.

The student noticed that surrounding the granite intrusion was a thin zone within the country rock where the sediments had been altered. Suggest a name for this thin zone.

End of Section Two

See next page

Section Three: Extended response **30% (30 Marks)**

This section contains **three (3)** questions. You must answer **two (2)** questions: the compulsory question (Question 25) and **one (1)** of the other questions (Question 26 or Question 27). Write your answers in the lined pages provided following Question 27.

If you use a page for planning, indicate this clearly at the top of the page.

Suggested working time: 60 minutes.

Compulsory question

Question 25 **(15 marks)**

A report published by the Australian Government Bureau of Meteorology and CSIRO in March 2012 describing changes in Australia's climate made the following key points:

- The average temperature in Australia has risen by 0.9°C since 1910.
- A shift in pressure systems associated with winter rainfall in southern Australia has led to a rapid decline in winter rainfall and extended periods of severe drought across most of the continent and a decline in average rainfall.
- The heavy summer rainfall and severe flooding experienced along the east coast of Australia in the summers of 2011 and 2012 were linked to changes in pressure and wind flow over the Pacific Ocean and rising sea water temperatures.
- The trend toward increasing average temperatures and prolonged periods of drought across most of Australia is likely to continue.

(a) Account for the rise in temperatures across Australia in the past 100 years. (5 marks)

(b) The report suggests that the long-term climatic trend for Australia is decreased average rainfall and increased drought periods. Despite this, record rainfall and severe flooding occurred in eastern Australia in 2011 and 2012. Explain this record rainfall and flooding. (5 marks)

(c) Suggest **three (3)** effects that increased temperatures and more frequent droughts may have on Australia's biodiversity. Illustrate your answer with possible examples. (5 marks)

Answer Question 26 or Question 27.

Question 26 **(15 marks)**

Western Australia's economy relies heavily on its natural resources (including oil/natural gas, iron ore, gold, fisheries and forestry).

Discuss how the extraction and processing of natural resources can lead to pollution and problems related to the sustainability of the resource. In your answer, you should:

(a) describe **two (2)** ways in which the extraction and processing of natural resources can lead to pollution. (4 marks)

(b) suggest **two (2)** pollution control methods that could be implemented to reduce the pollution described in Part (a). (4 marks)

(c) define what is meant by the term 'sustainability'. (3 marks)

(d) use a case study to discuss the ecological sustainability of a resource site. (4 marks)

or

Question 27 **(15 marks)**

Metamorphism has a very important role in the development of the Earth's crust. It produces rocks such as slate, schist, gneiss, marble, quartzite, hornfels and amphibolite. It also plays a part in the formation of many ore deposits.

Discuss the process of metamorphism and the rocks it produces. In your answer, you should:

(a) name and outline similarities and differences between the two dominant types of metamorphism. (6 marks)

(b) describe the textural and/or mineralogical features that allow metamorphic rocks to be distinguished from the other two rock groups (sedimentary and igneous). (4 marks)

(c) explain what is meant by the term 'metamorphic grade' and describe how the mineralogy and textures in a siltstone would change with increasing metamorphic grade. (5 marks)

End of questions

Question number: ____________

Question number: ______________

Question number: ______________

Question number:

Question number: ______________

Additional working space: ____________

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Additional working space: ______________

You may tear along the perforations to use this page (to transcribe strata locations for Question 24).
Insert this page back into the Question/Answer Booklet once you have completed Question 24.

This page is to be used for transcribing strata locations only

You may tear along the perforations to use this page (to transcribe strata locations for Question 24).
Insert this page back into the Question/Answer Booklet once you have completed Question 24.

This page is to be used for transcribing strata locations only

Question 21 spare grid.

ACKNOWLEDGEMENTS

Section Two

Question 20	Adapted from data source: Kasting, J.F. (1993). Earth's early atmosphere. *Science*, 259, pp. 920–926. Retrieved March, 2012, from http://volcano.oregonstate.edu/education/gases/origin.html.
Question 21	Data source: National Aeronautics and Space Administration. Goddard Space Flight Center. (n.d.). *Ozone hole watch.* Retrieved March, 2012, from http://ozonewatch.gsfc.nasa.gov/meteorology/annual_data.html.
Question 23(c)	Adapted from: GeoMan. (n.d.). *Bowen's reaction series* [Image]. Retrieved March, 2012, from http://jersey.uoregon.edu/~mstrick/AskGeoMan/geoQuerry32.html.
Question 24	Geological map by courtesy of the examining panel.

Section Three

Question 25	Data source: Bureau of Meteorology. (2012). *Media release: State of the climate 2012: Australia continues to warm.* [Melbourne]: Bureau of Meteorology. Retrieved March, 2012, from www.bom.gov.au/.

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